



HEMOSTASIS

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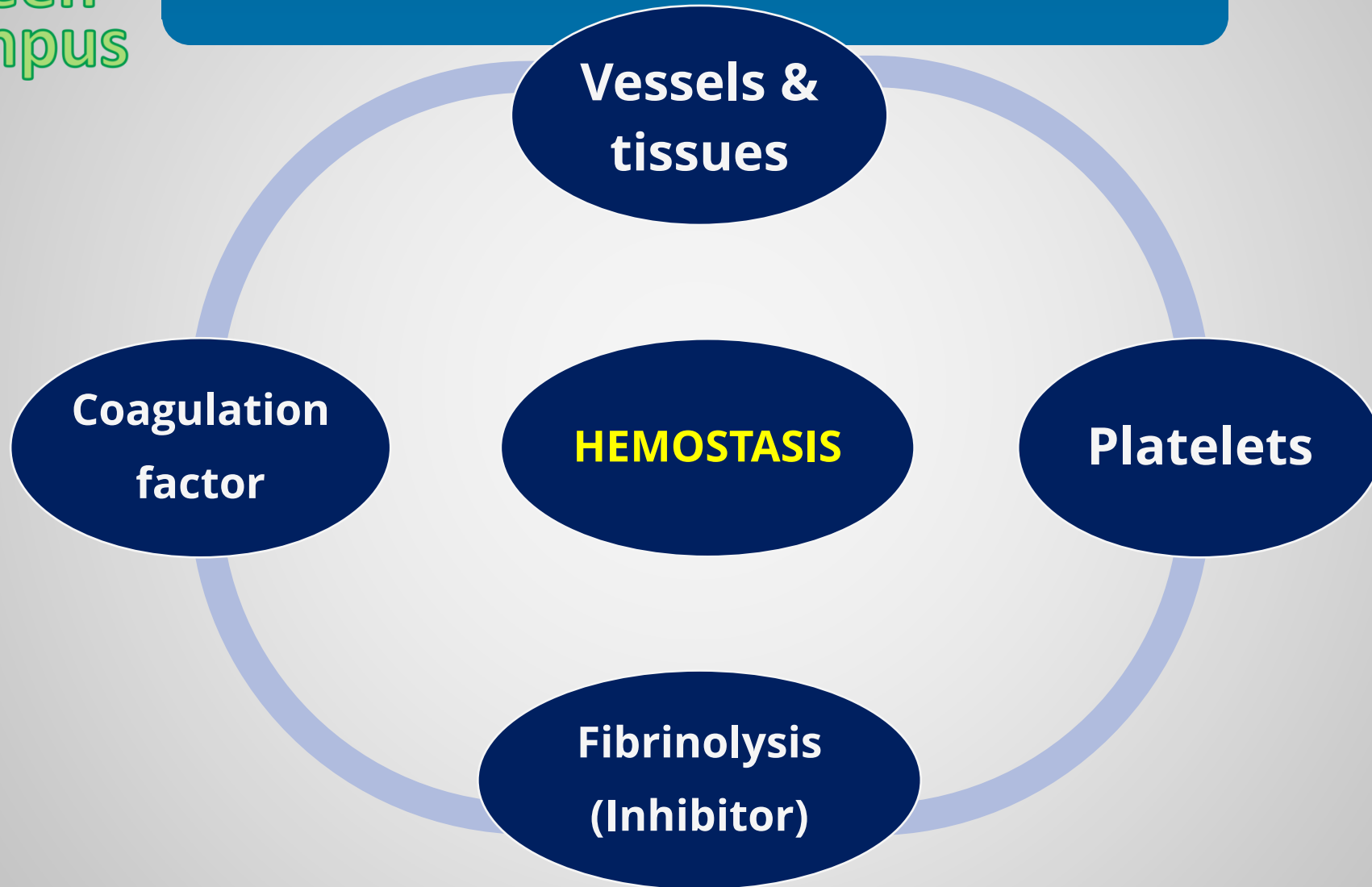


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Homeostasis of Hemostasis



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MECHANISM OF HEMOSTASIS

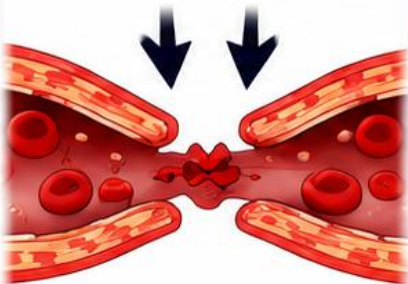
Hemostasis is the physiologic process that stops bleeding at the site of vascular injury and maintains blood in a fluid state within vessels.

OVERVIEW OF HEMOSTASIS

1 VASCULAR SPASM

(Immediate)

Transient vasoconstriction reduces blood flow and blood loss.



2 PLATELET PLUG FORMATION

(Primary Hemostasis)

Platelets adhere, activate, and aggregate to form a temporary plug.



3 COAGULATION

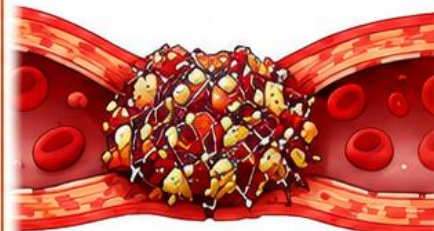
(Secondary Hemostasis)

Coagulation cascade produces fibrin, which stabilizes the platelet plug into a strong clot.



4 CLOT RETRACTION & STABILIZATION

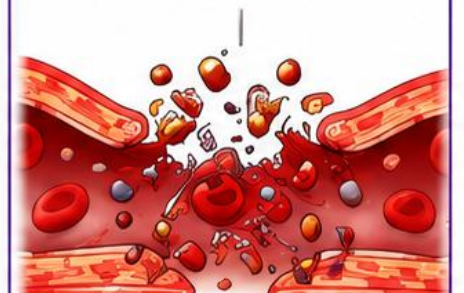
Platelet actin–myosin contraction retracts the clot and strengthens it.



5 FIBRINOLYSIS

(Clot Removal)

Clot is degraded by fibrinolysis after healing to restore normal blood flow.

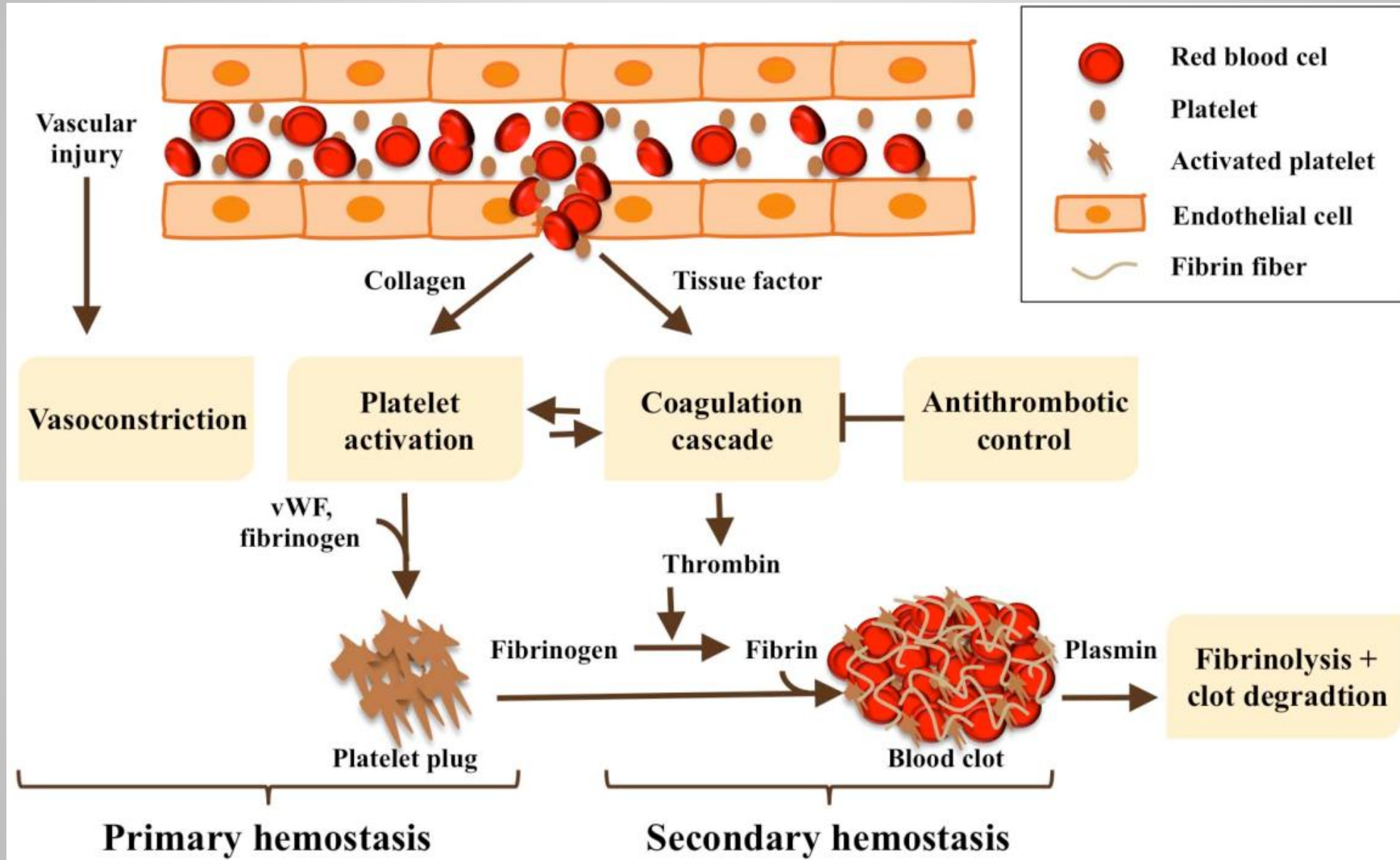


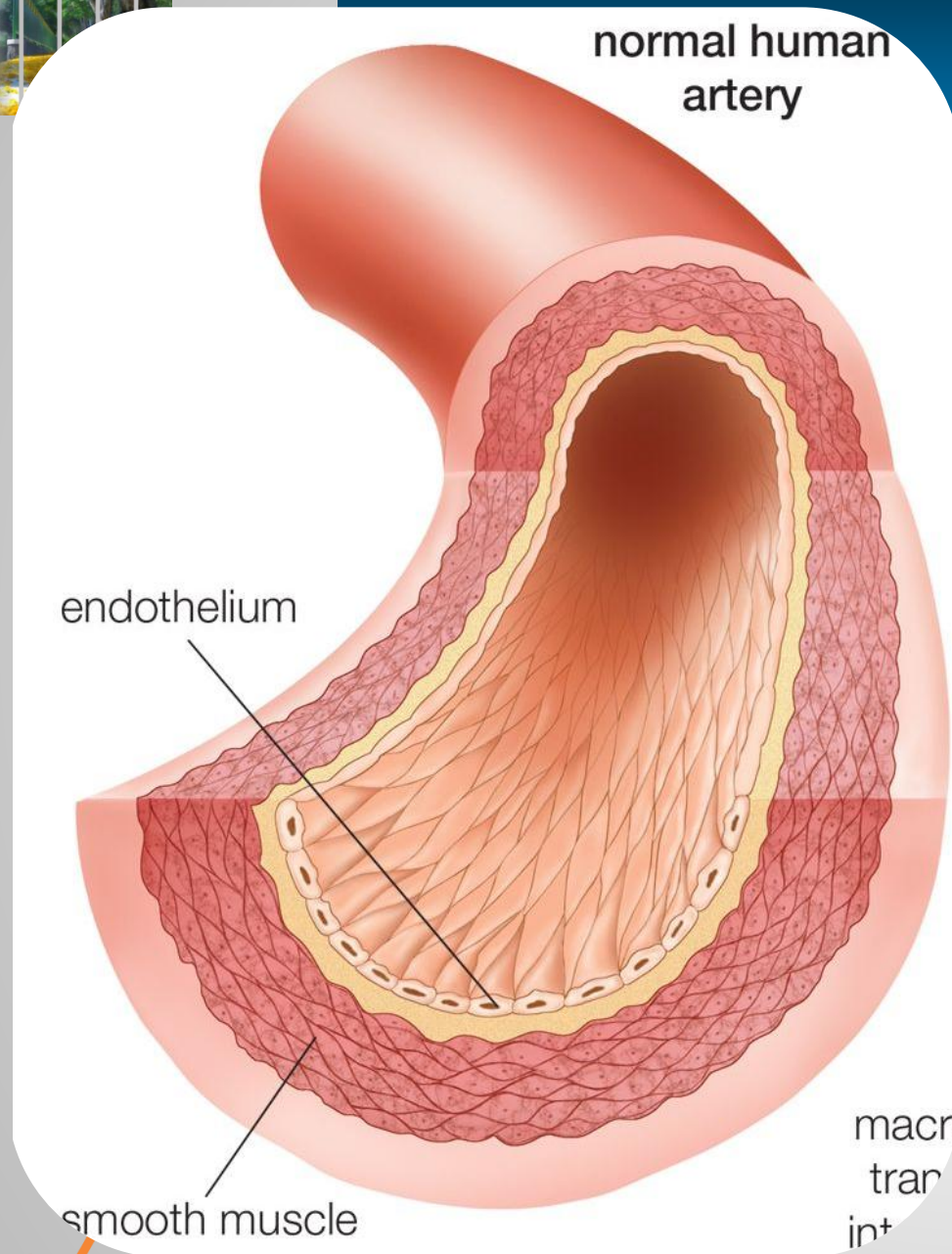
Primary Hemostasis

Secondary Hemostasis

Tertiary Hemostasis

MAJOR COMPONENTS OF HEMOTASIS





BLOOD VESSEL

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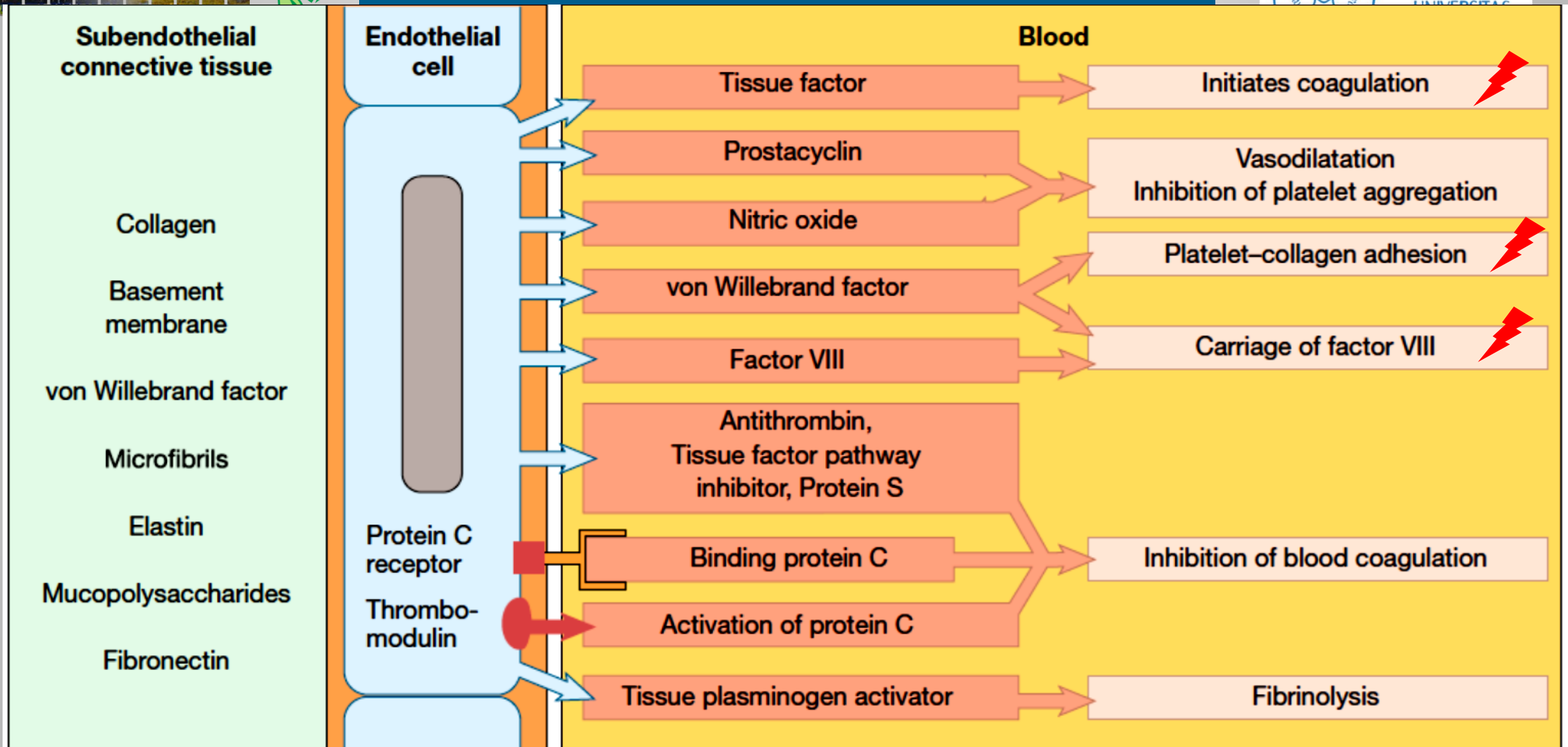
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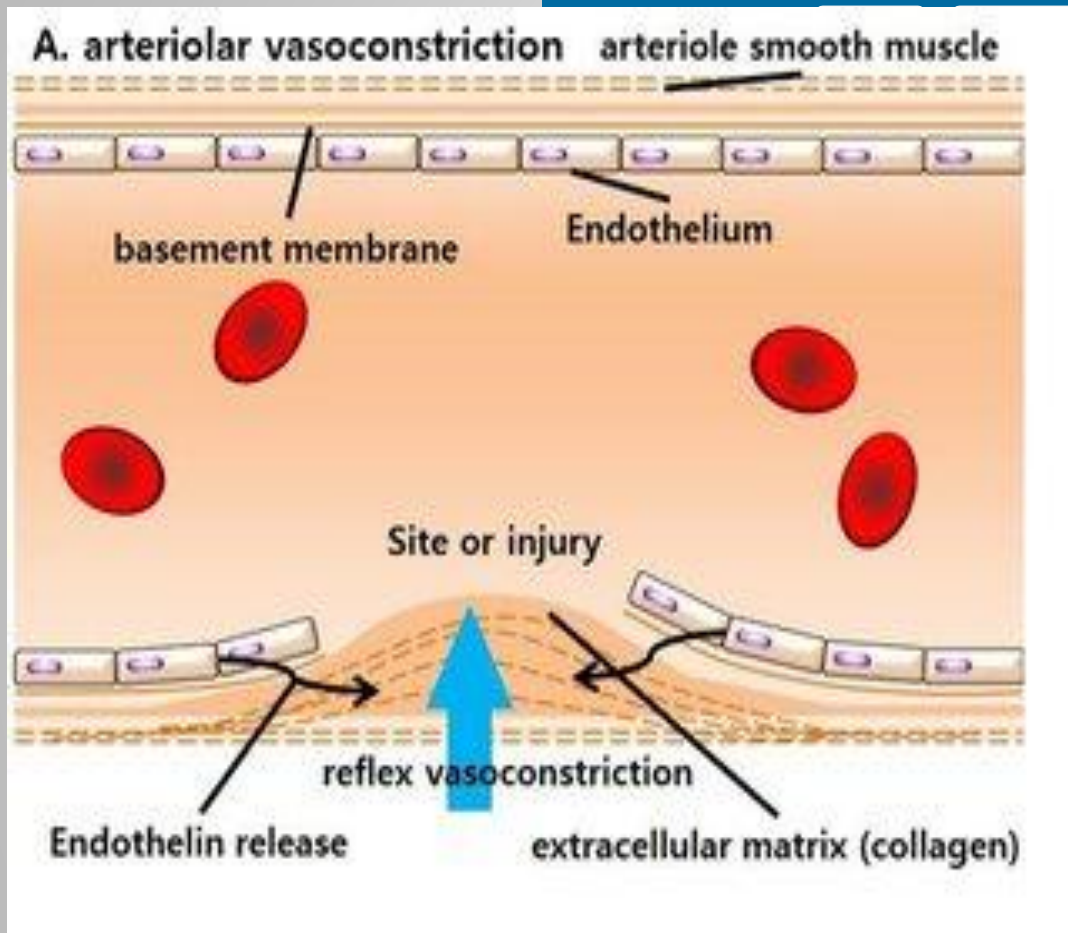

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Blood Vessel





Blood Vessels



Endothel broken → endothelin. Effect:

- vasoconstriction
- with trombin induce adhesion substance (integrin & selectine)
- attract leucocyte & thrombocyte to injured endothel

Endothel can be injured by :

- Acidosis
- Hipoxia
- Toxin
- Antigen-antibody complex
- High blood pressure,

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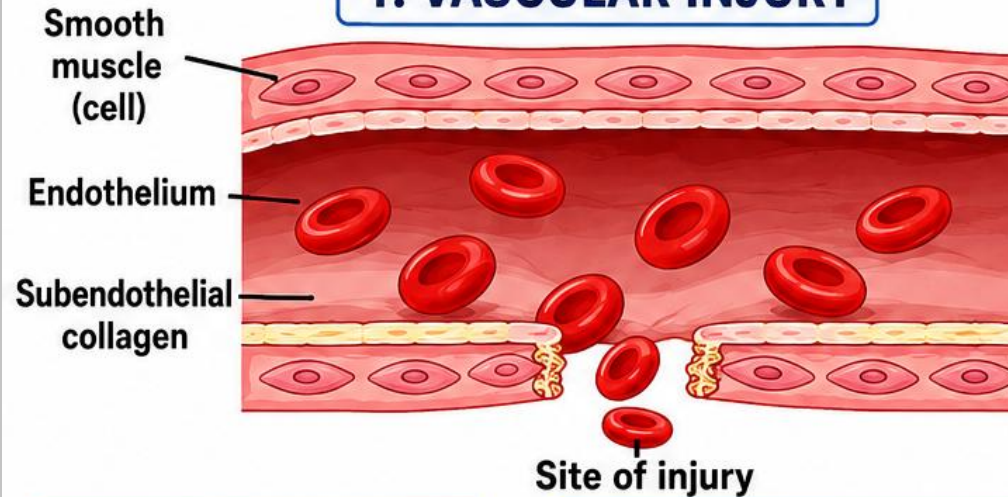
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HEMOSTASIS: VASCULAR SPASM PHASE

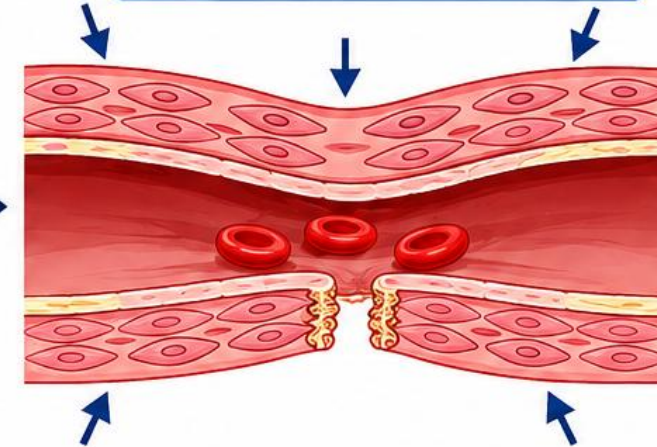
Immediately after vascular injury, the affected blood vessel constricts to reduce blood flow and limit blood loss.

MECHANISM

1. VASCULAR INJURY



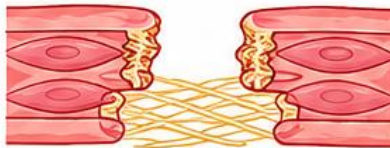
2. VASCULAR SPASM



RESULT:

Reduced blood flow at the injury site and limited blood loss

- 1** Vascular injury exposes the subendothelial tissue.

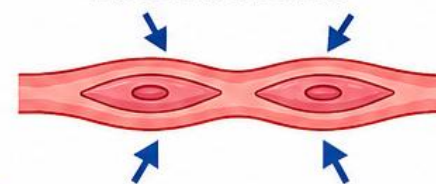


- 2** Endothelial cells and platelets release vasoconstrictors:

- Endothelin
- Thromboxane A₂ (TXA₂)
- Serotonin
- ADP



- 3** These mediators act on vascular smooth muscle cells, causing vasoconstriction.



- 4** Vasoconstriction reduces blood flow to the injured area, promoting the next hemostatic steps (platelet adhesion).

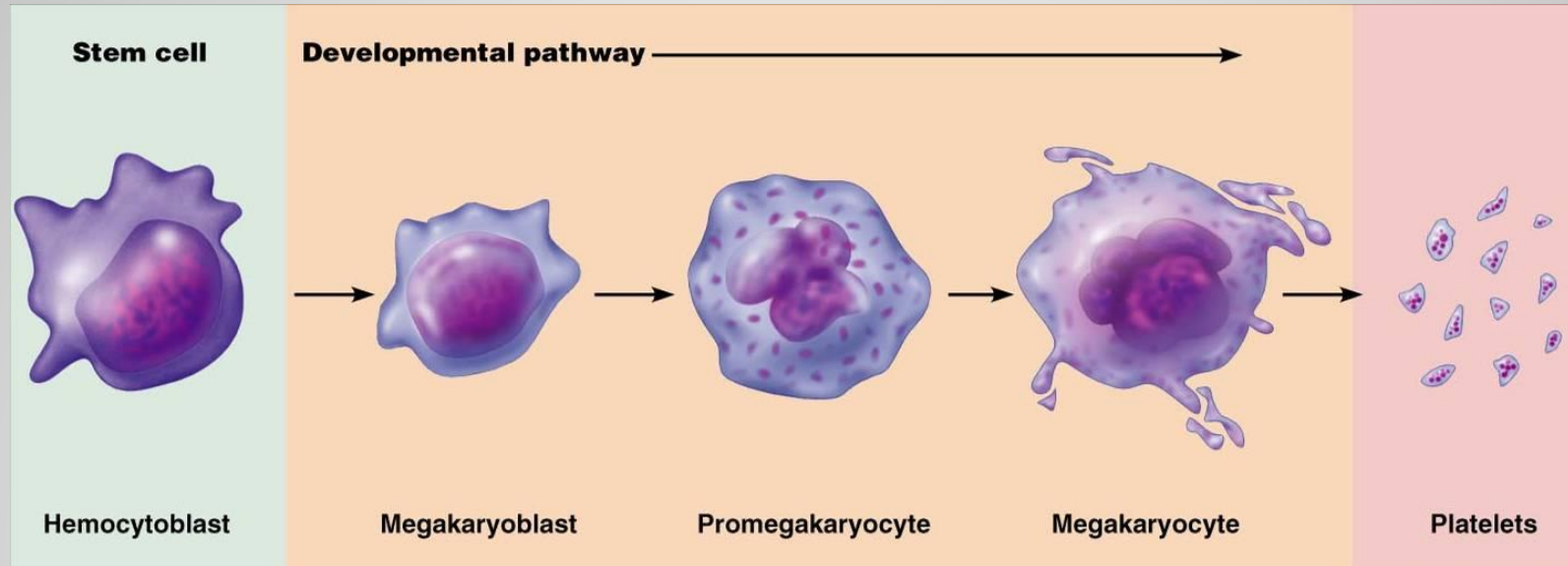


KEY POINT: Vascular spasm is a rapid, transient response that occurs within seconds to minutes after injury, helping to minimize blood loss and creating a favorable environment for platelet plug formation.

Vascular Phase

1. Immediate **reflex vasoconstriction** : Intrinsic response of smooth muscle in small arterioles & pre capillary sphincters
2. Sustained vascular contraction : due to release of local autocooid factors from traumatized tissue & platelets (ex : TXA2, epinephrine, fibrinopeptide B)

Thrombocytes - Platelets



- Megakaryocytes shed small cytoplasmic fragments
- $150,000-400,000/\text{mm}^3$
- Platelets circulate in a resting, **inactive state**. On stimulation, undergo major shape change, develop receptors for clotting factors, develop ability to bind to each other & subendothelium
- Life-span: 7~14 days in the blood
- Breakage: Aged platelet can be processed by phagocytosis in liver, spleen and lymphatic node



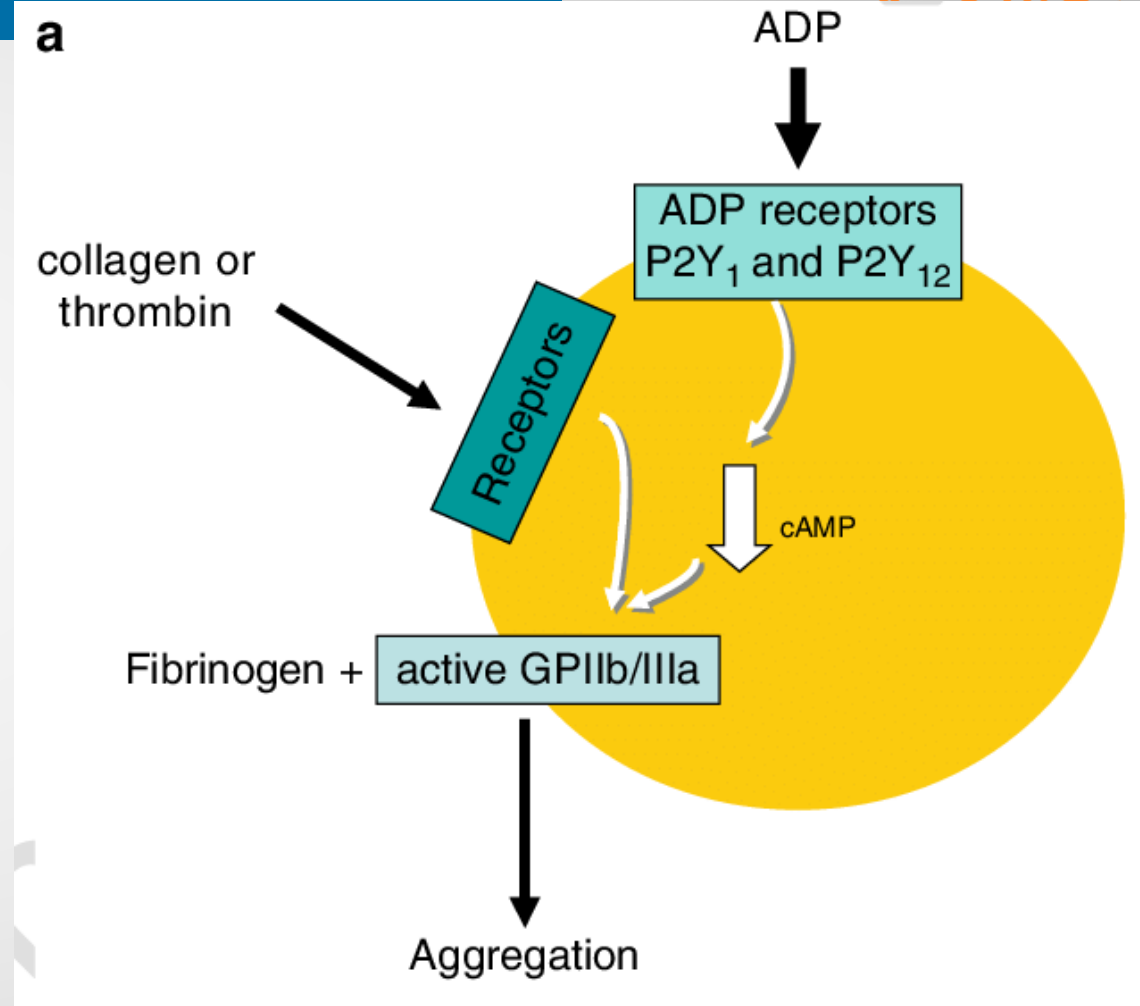
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Factors that activate platelets



Thrombin
Collagen
Adenosine Diphosphat (ADP)
5-hydroxytryptamine (serotonin)
Thromboxane A₂
Mechanical stimuli

Many stimuli
Several different receptors
Multiple signalling pathways



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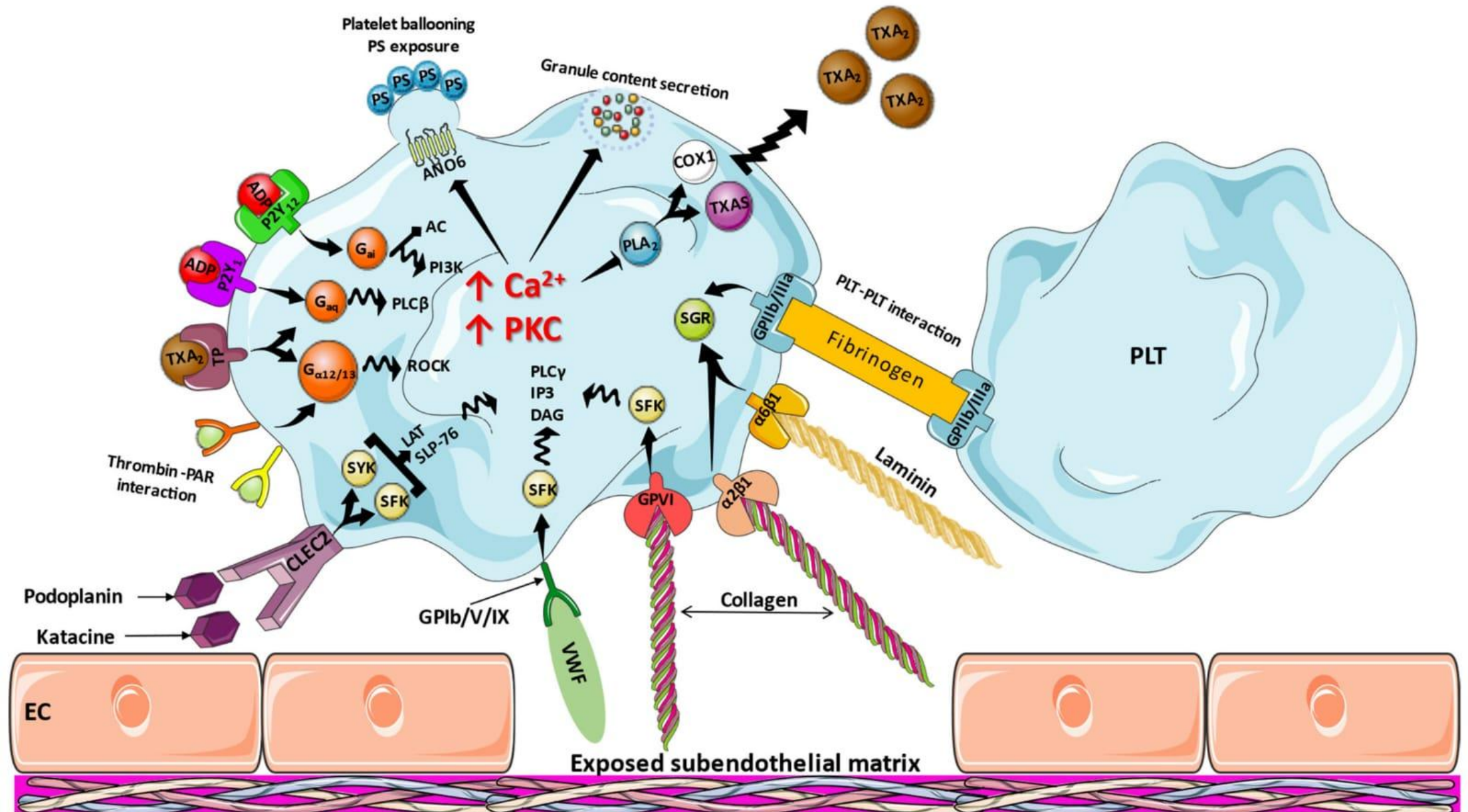
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Platelet Plug Formation

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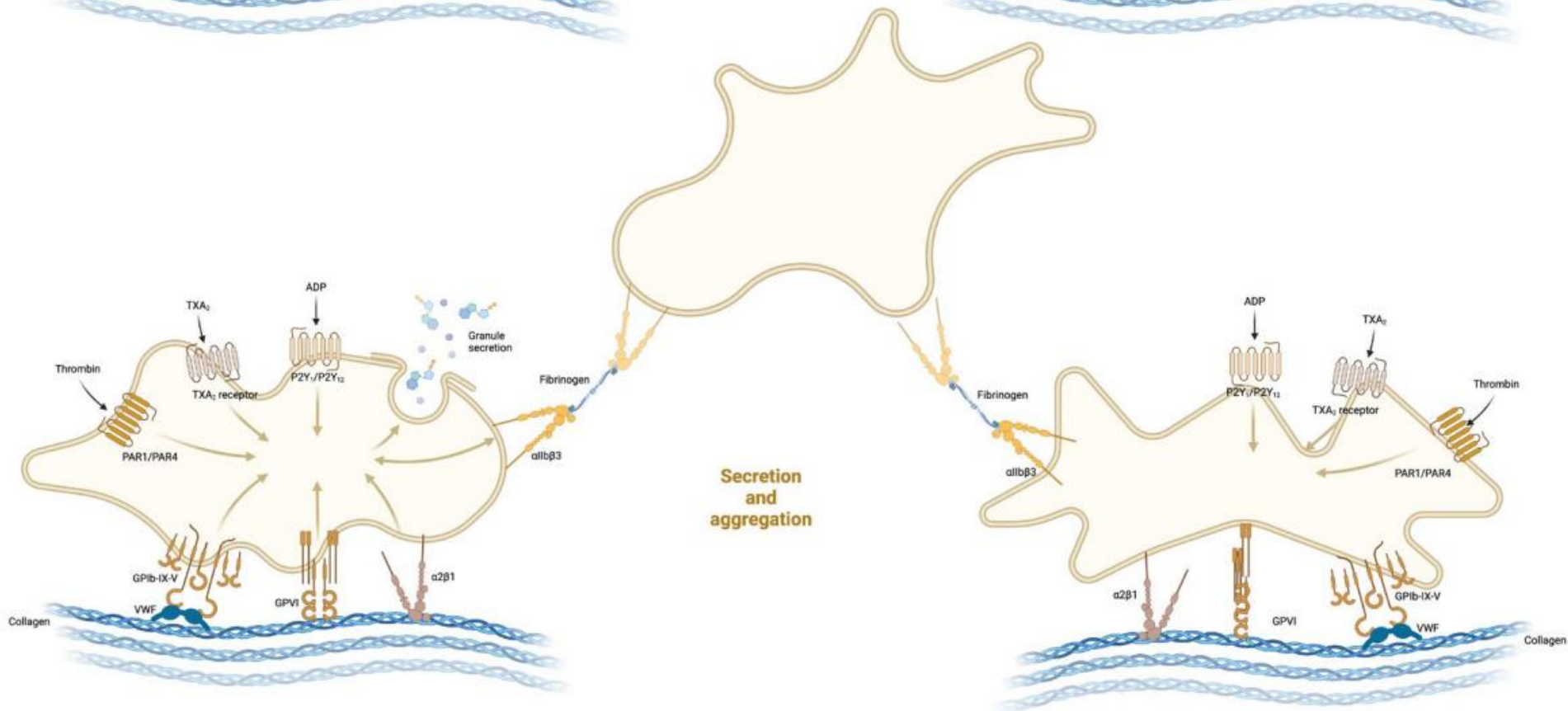
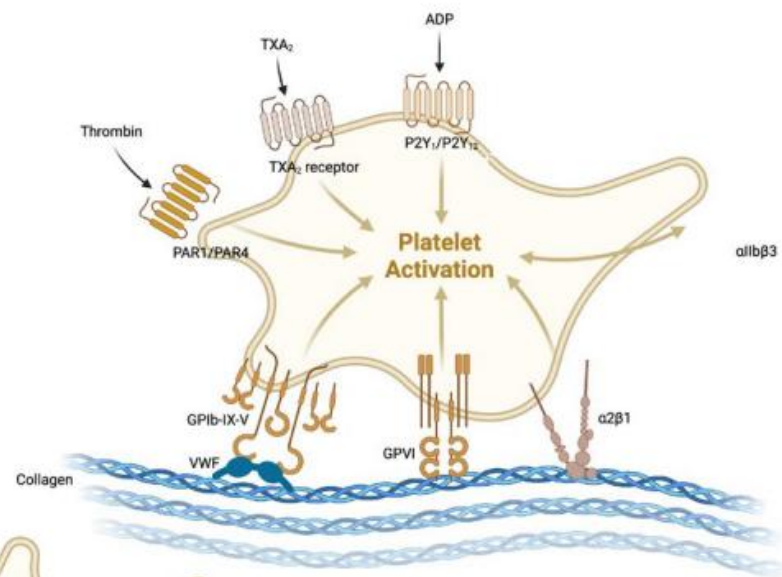
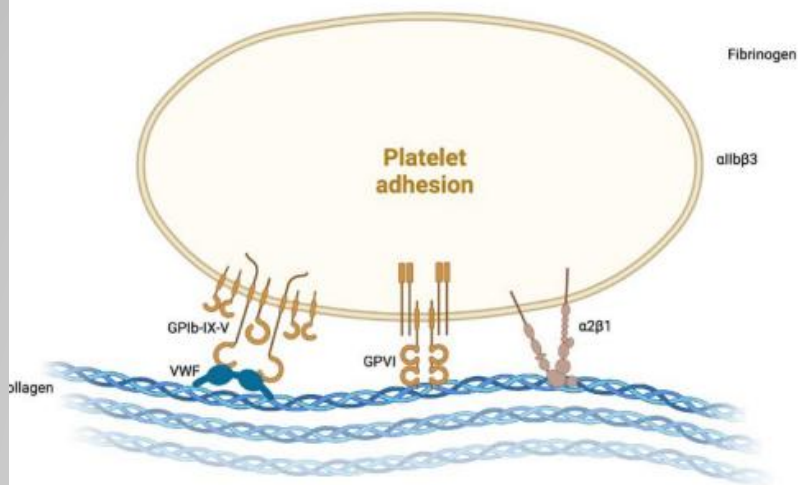


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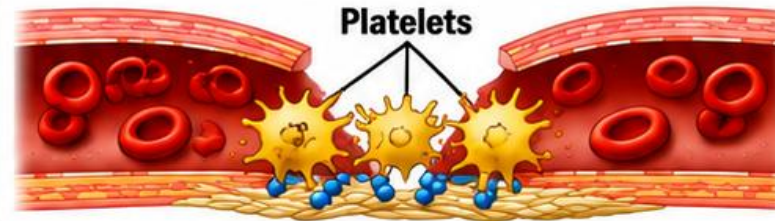


MECHANISM OF PLATELET PLUG FORMATION (Primary Hemostasis)



2 PLATELET ADHESION

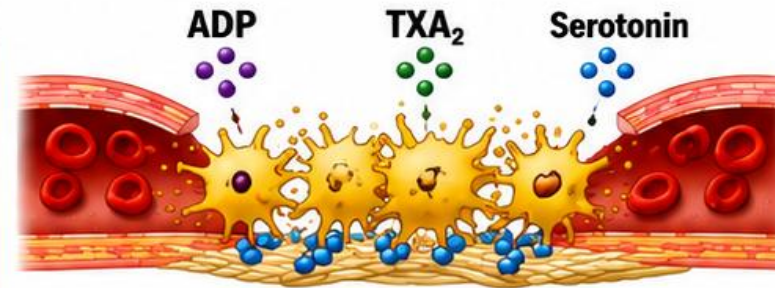
- Platelets adhere to exposed collagen through vWF.
- The GPIb receptor on platelets binds to vWF.



Key mediator/receptor:
vWF – GPIb interaction

3 PLATELET ACTIVATION

- Adherent platelets become activated.
- Shape change occurs and granule contents are released.

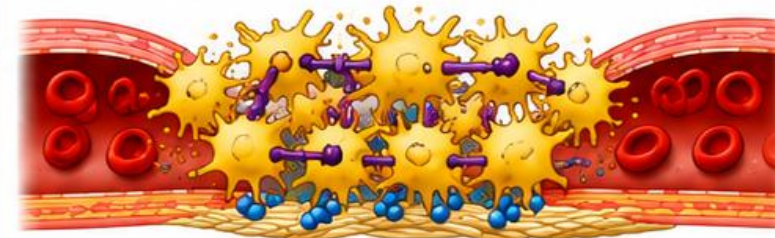


Released mediators:
ADP, TXA₂ (thromboxane A₂),
Serotonin, Ca²⁺

Functions:
Recruit more platelets and
amplify activation

4 PLATELET AGGREGATION

- Activated platelets express GPIIb/IIIa receptors.
- Fibrinogen binds between GPIIb/IIIa receptors on platelets (bridging).



Key mediator:
Fibrinogen bridges GPIIb/IIIa
receptors

Result:
Platelets aggregate to form
a temporary plug

5 PLATELET PLUG FORMATION

- A loose platelet plug is formed to stop bleeding temporarily.
- This plug will be stabilized by fibrin (secondary hemostasis).



Outcome:
Temporary platelet plug
(Primary hemostasis)
prevents further blood loss
until fibrin clot stabilizes it.

LEGEND



Red blood cell



Platelet
(resting)



Platelet
(activated)

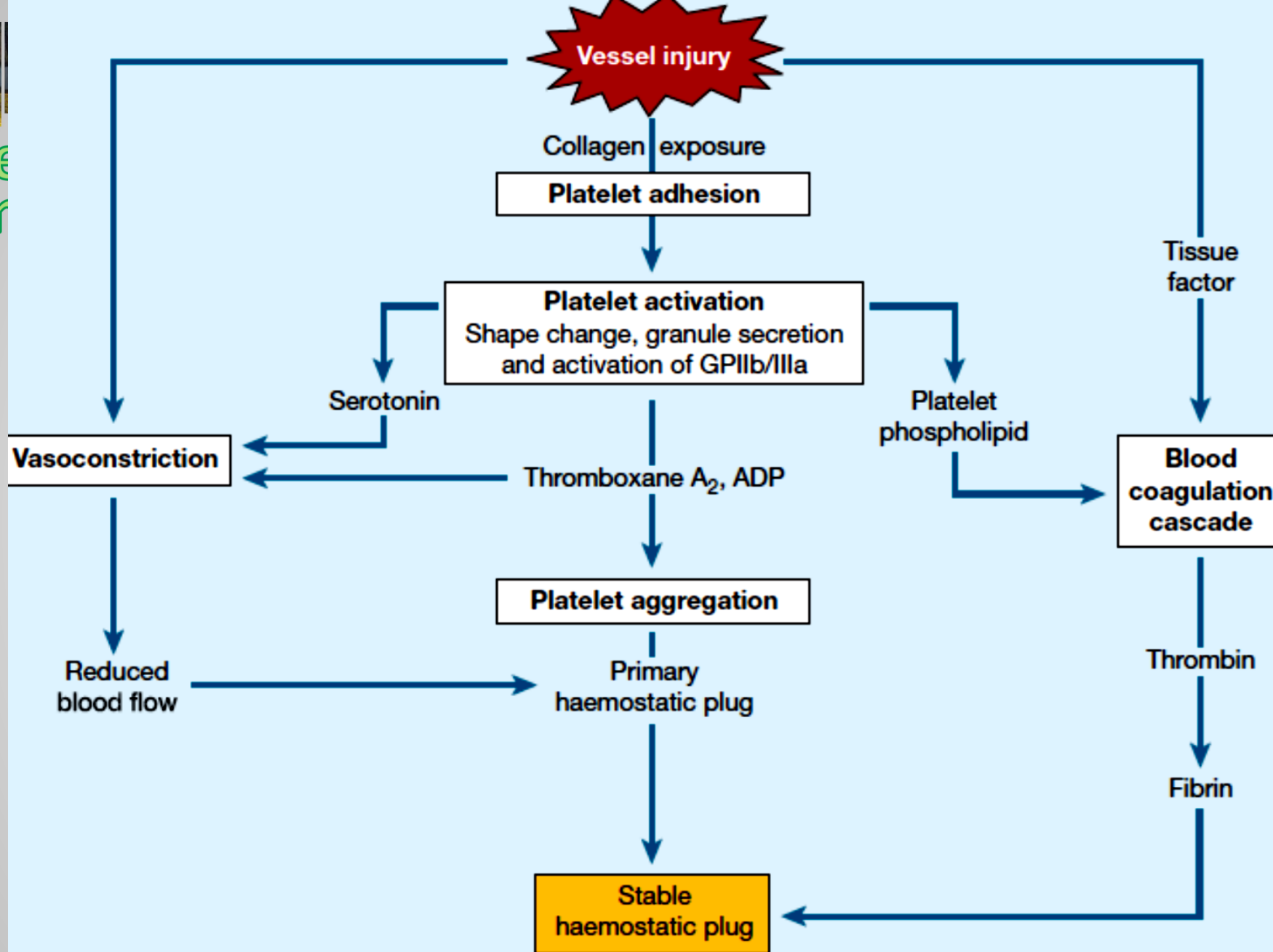
von Willebrand
factor (vWF)



Collagen



Fibrinogen





BLOOD COAGULATION

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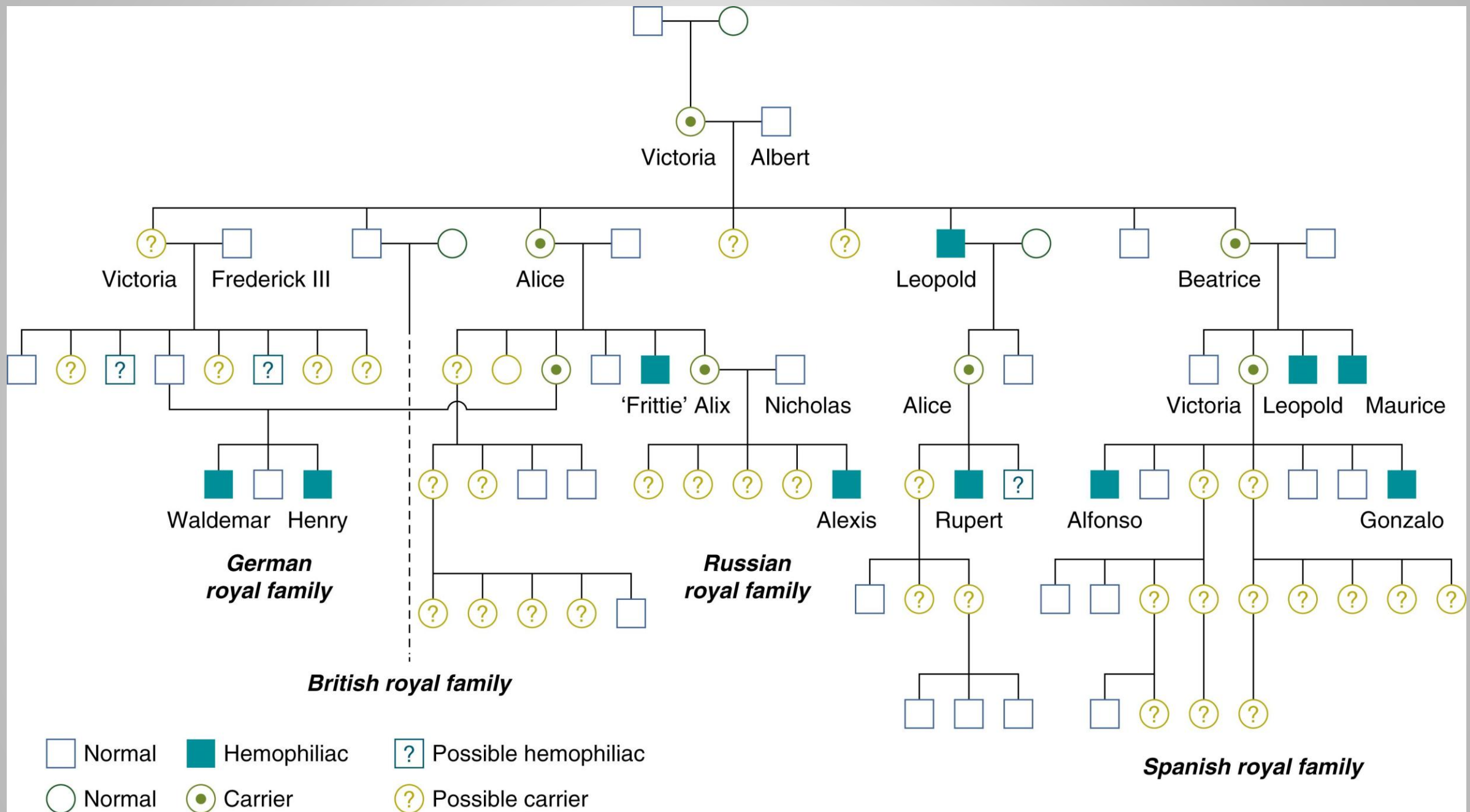


Table 24.1 The coagulation factors.

Factor number	Descriptive name	Active form
I	Fibrinogen	Fibrin subunit
II	Prothrombin	Serine protease
III	Tissue factor	Receptor/cofactor*
V	Labile factor	Cofactor
VII	Proconvertin	Serine protease
VIII	Antihaemophilic factor	Cofactor
IX	Christmas factor	Serine protease
X	Stuart–Prower factor	Serine protease
XI	Plasma thromboplastin antecedent	Serine protease
XII	Hageman (contact) factor	Serine protease
XIII	Fibrin-stabilizing factor	Transglutaminase
	Prekallikrein (Fletcher factor)	Serine protease
	HMWK (Fitzgerald factor)	Cofactor*

HMWK, high molecular weight kininogen.

* Active without proteolytic modification.

Active!

The Coagulation Factors

The three pathways that make up the classical blood coagulation pathway

Intrinsic

surface contact

XII → XII_a

XI → XI_a

IX → IX_a

X (VIII, PL, Ca⁺⁺) → X_a

prothrombin (V, PL, Ca⁺⁺) → thrombin (serine protease)

fibrinogen

→ fibrin

→ XIII_a

stable fibrin clot

XII – Hageman factor, a serine protease

XI – Plasma thromboplastin, antecedent serine protease

IX – Christmas factor, serine protease

VII – Stable factor, serine protease

XIII – Fibrin stabilising factor, a transglutaminase

PL – Platelet membrane phospholipid

Ca⁺⁺ – Calcium ions

TF – Tissue Factor

(_a = active form)

Extrinsic

TF:VII_a ← tissue damage

Common

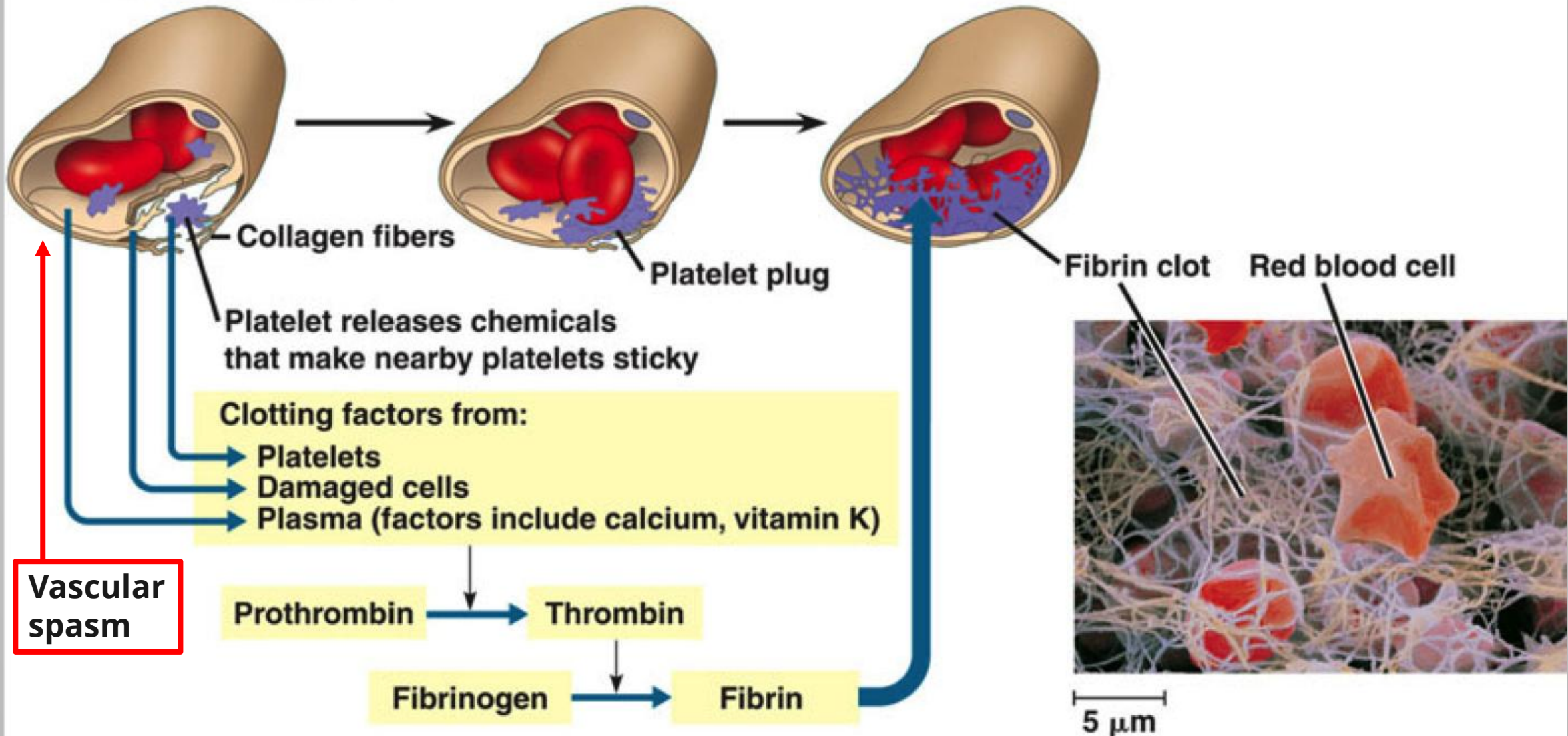
X → X_a

thrombin (serine protease)

XIII

→ XIII_a

- 1 Endothelium of vessel is damaged, exposing connective tissue; platelets adhere
 - 2 Platelets form a plug
 - 3 Seal is reinforced by a clot of fibrin
- coagulation**



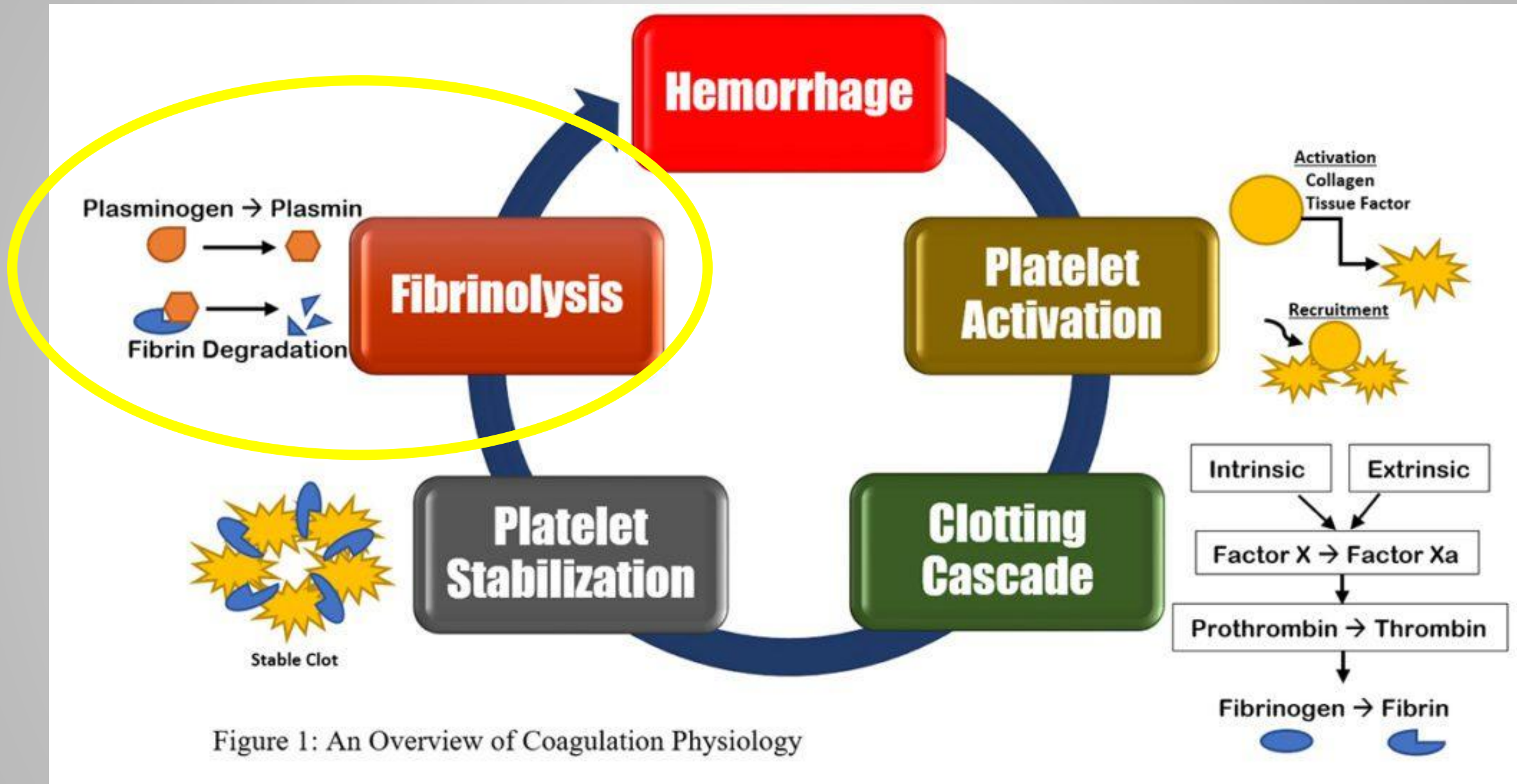


Figure 1: An Overview of Coagulation Physiology

Blood Coagulation and Fibrinolysis

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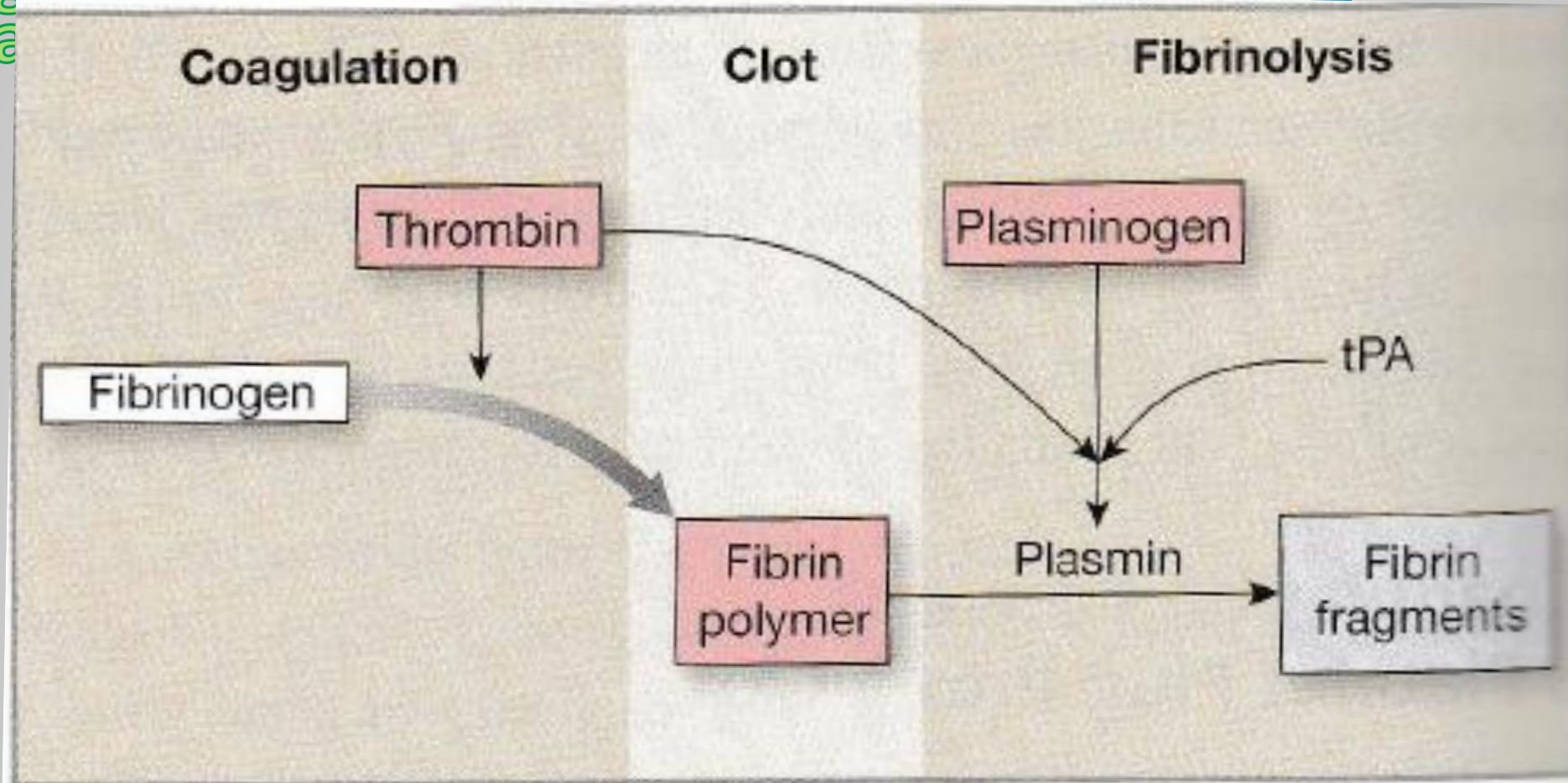


FIGURE 16-13 *Coagulation and fibrinolysis*

Fibrinolysis

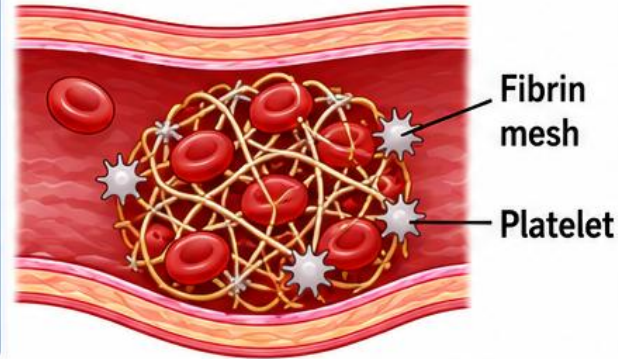
- Fibrinolysis - dissolution of a clot, begins within 2 days
 - plasminogen trapped in clot
 - factors convert **plasminogen** into **plasmin (fibrinolysin)** :
 - Thrombin
 - activated factor XII
 - tissue plasminogen activator (t-PA)
- Plasmin
 - enzymatically digests fibrin threads
 - digests fibrinogen, prothrombin, and several clotting factors

MECHANISM OF FIBRINOLYSIS

The Body's Natural Process to Dissolve Blood Clots

1 FIBRIN CLOT FORMATION

After vascular injury, the coagulation cascade forms a fibrin clot to stop bleeding.



2 ACTIVATION OF PLASMINOGEN

Endothelial cells release tissue-type plasminogen activator (tPA).

tPA binds to fibrin in the clot and converts plasminogen to plasmin.

Plasminogen
(inactive)

tPA

Plasmin
(active)

Source of tPA



Endothelial cells (vascular lining) release tPA into the circulation

3 ACTION OF PLASMIN

Plasmin is a serine protease that cleaves fibrin strands in the clot.



FDPs include:

- D-dimer
- Degradation fragments

4 INHIBITION AND REGULATION

The body tightly regulates fibrinolysis to prevent excessive bleeding.

PAI-1

(Plasminogen Activator Inhibitor-1)

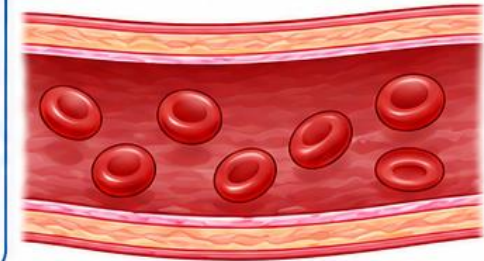
Inhibits tPA and prevents the conversion of plasminogen to plasmin.

α_2 -Antiplasmin

Binds to plasmin and inhibits its activity, preventing further fibrin degradation.

5 RESULT

The clot is gradually degraded and removed, restoring normal blood flow.



LEGEND



= Activation / Conversion

= Inhibition

tPA

= Tissue-type plasminogen activator

PAI-1

= Plasminogen activator inhibitor-1

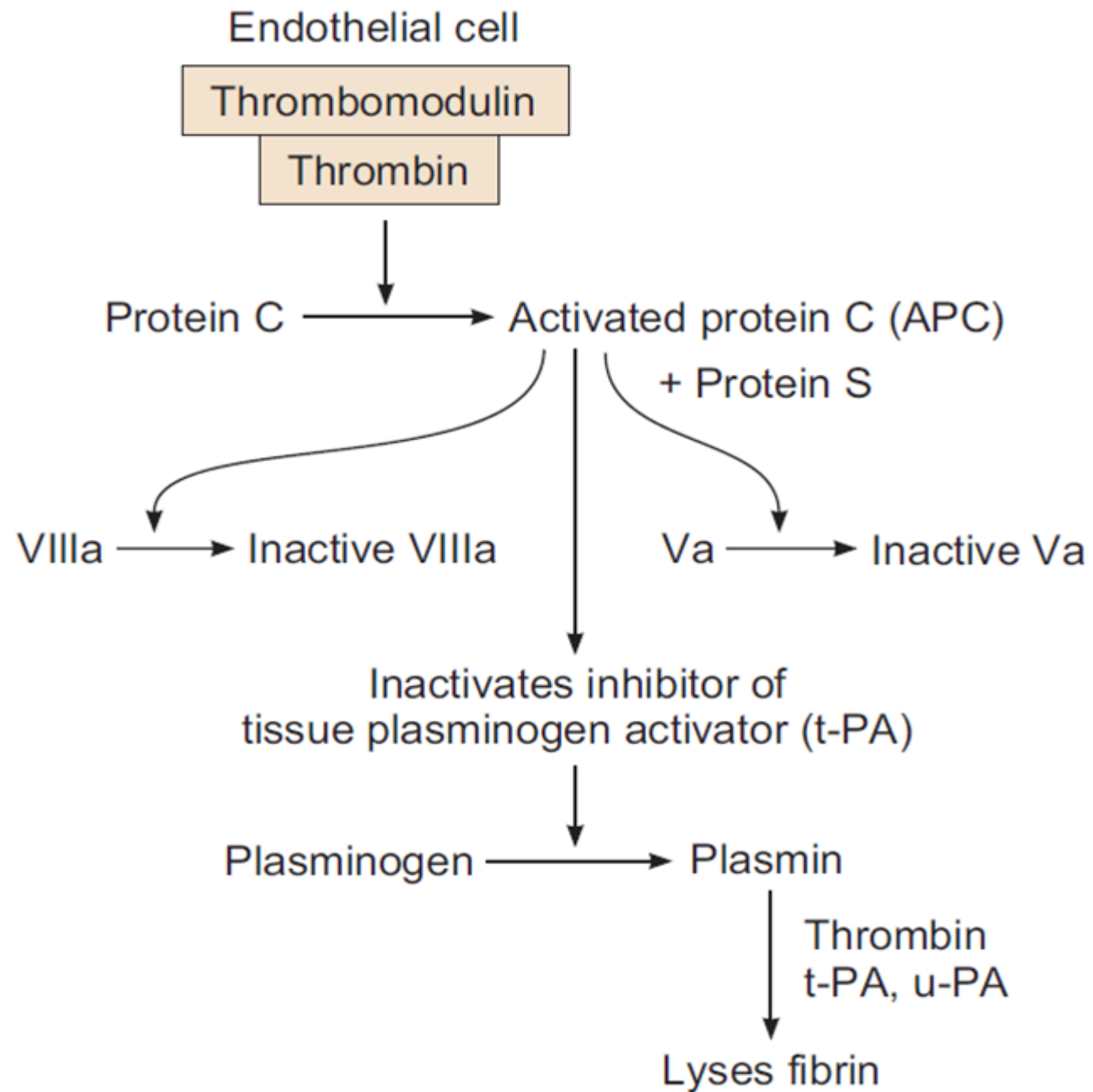
FDPs

= Fibrin degradation products

α_2 -Antiplasmin

= Plasmin inhibitor

Fibrinolysis is the natural process that dissolves fibrin clots through the conversion of plasminogen to plasmin, leading to fibrin degradation and restoration of normal blood flow.





clinical implementation

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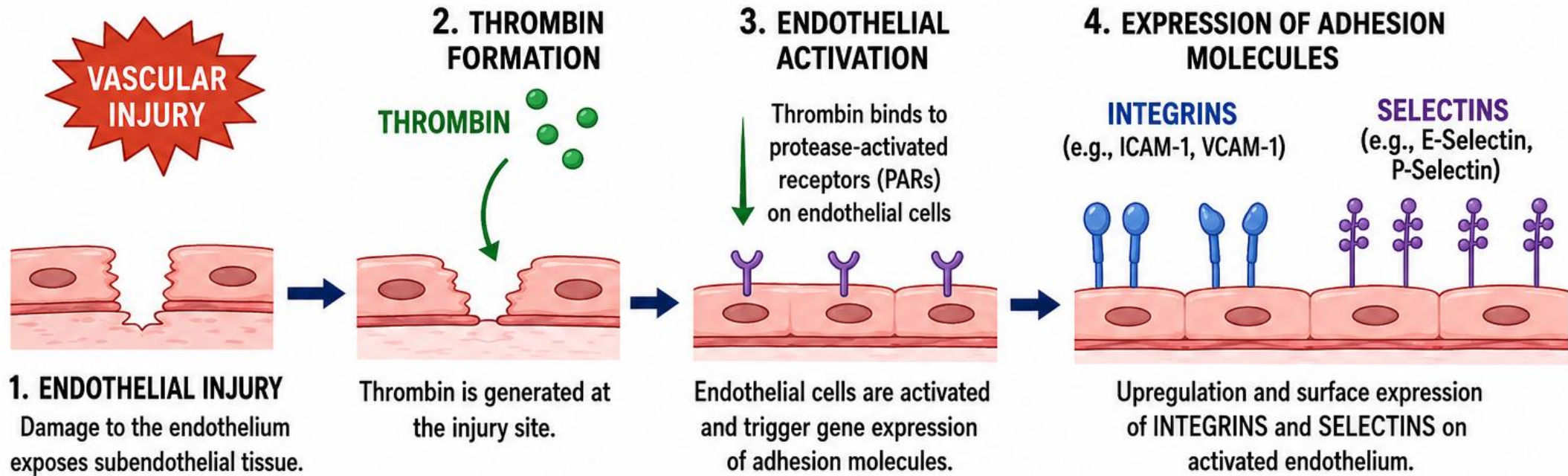
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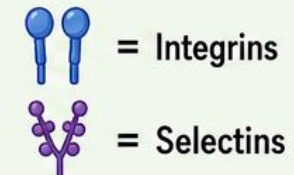
THROMBIN-INDUCED ADHESION SUBSTANCES (INTEGRINS & SELECTINS) EXPRESSION ON INJURED ENDOTHELIUM

Thrombin generated at the injury site activates endothelial cells to express adhesion molecules: **INTEGRINS** and **SELECTINS**

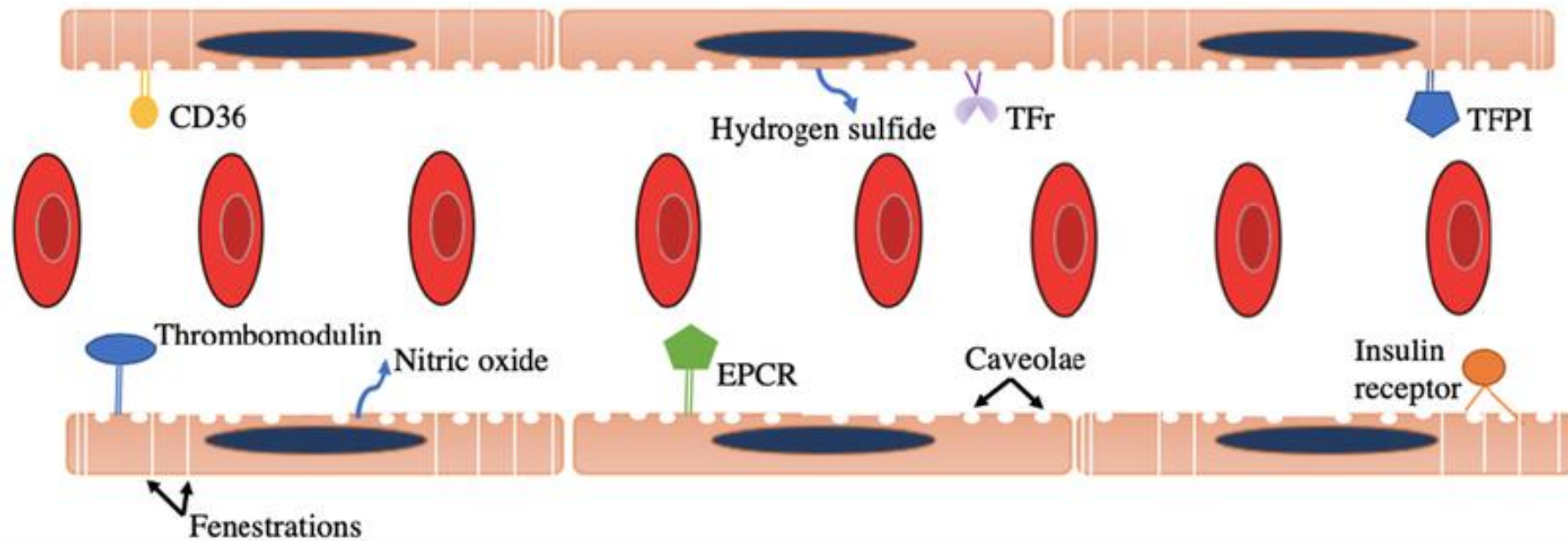


FUNCTION

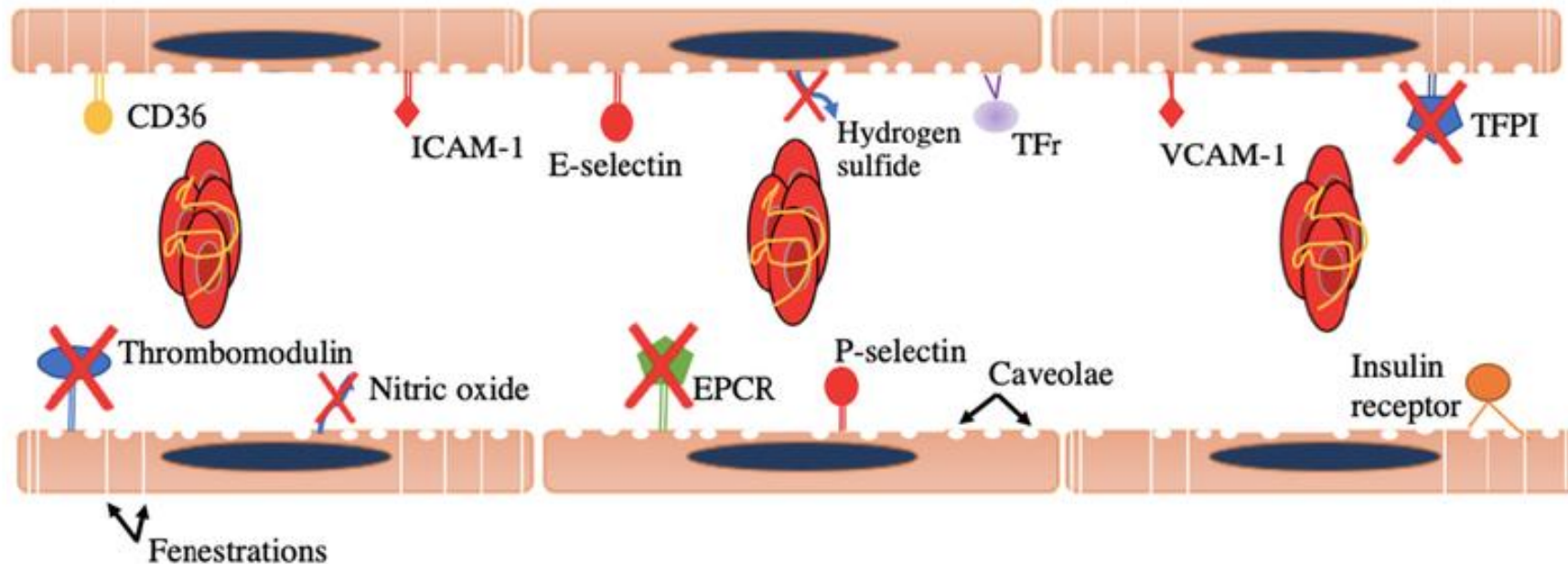
These adhesion molecules mediate the recruitment, rolling, adhesion, and transmigration of leukocytes to the site of injury, promoting inflammation and tissue repair.



A. Healthy endothelium

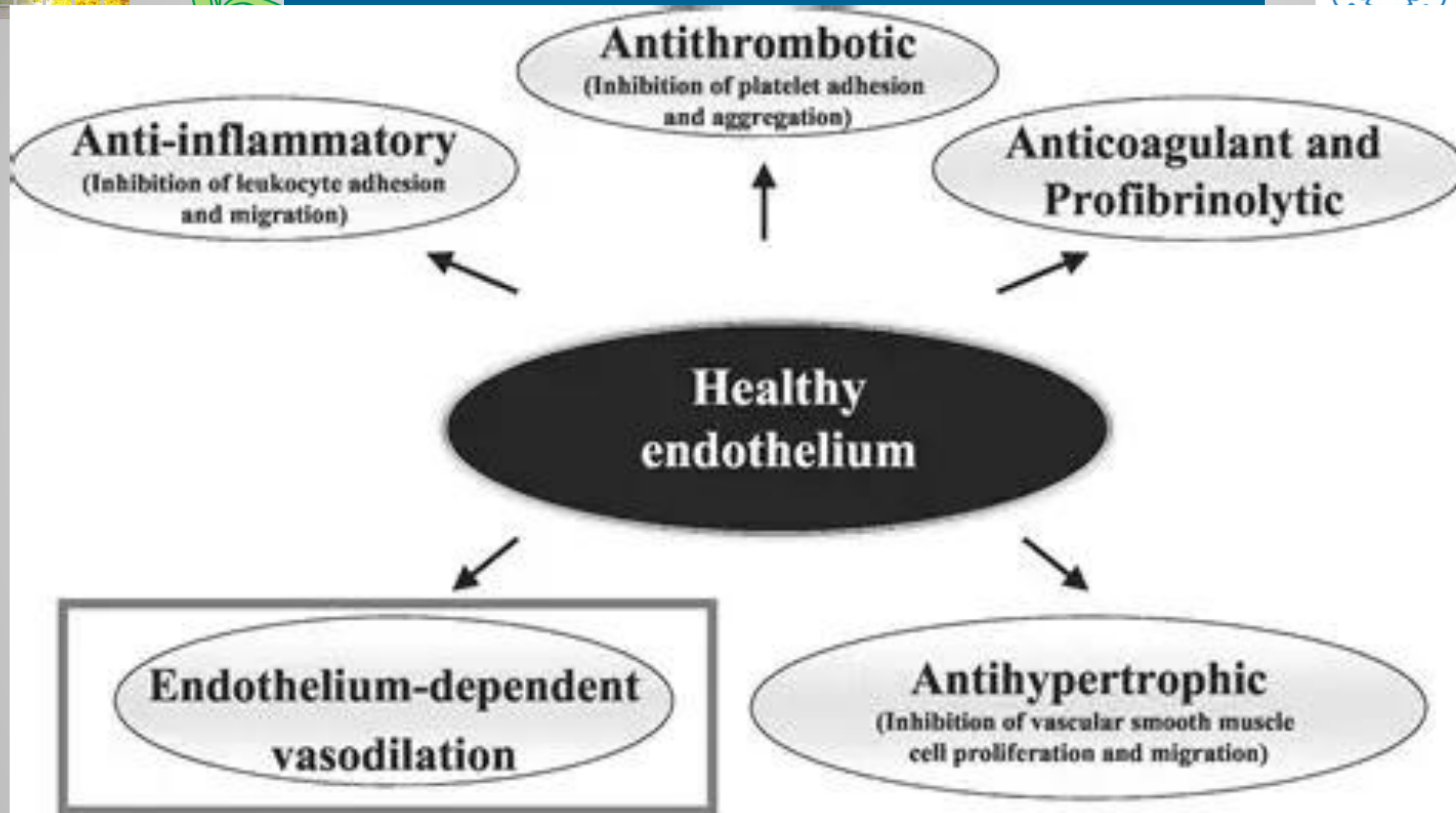


B. Diseased endothelium

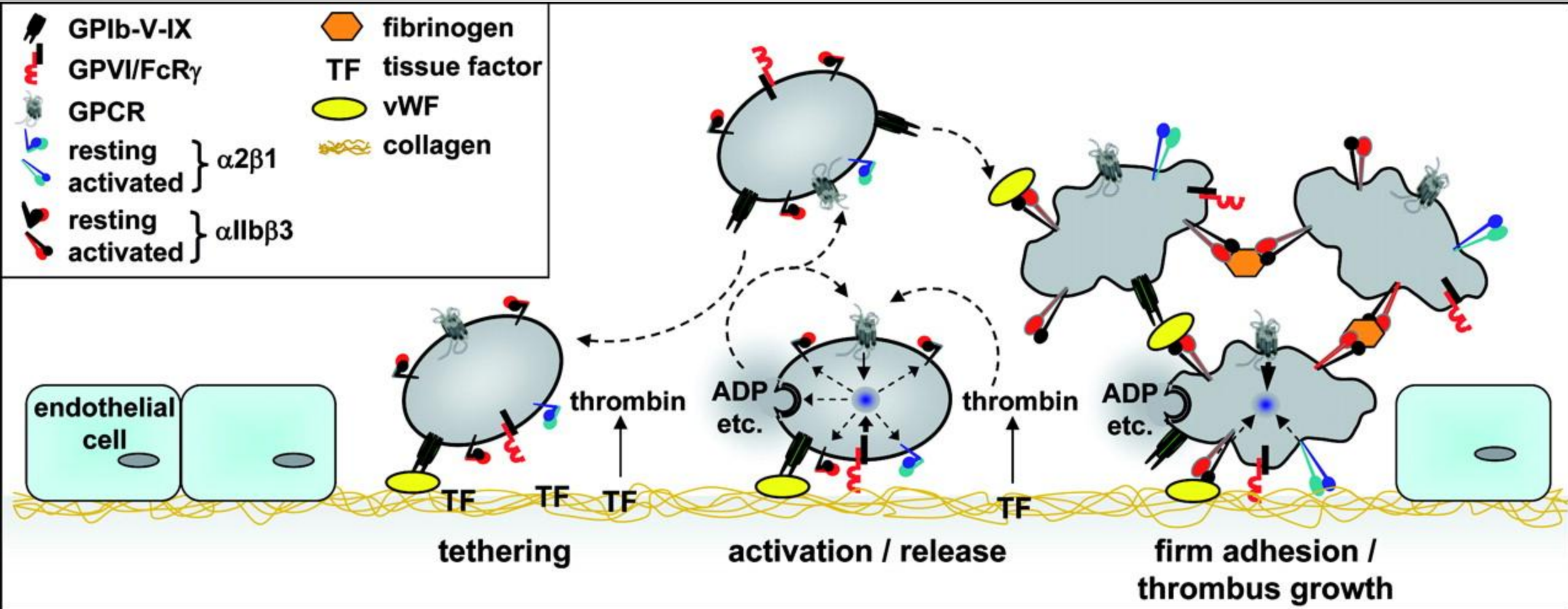


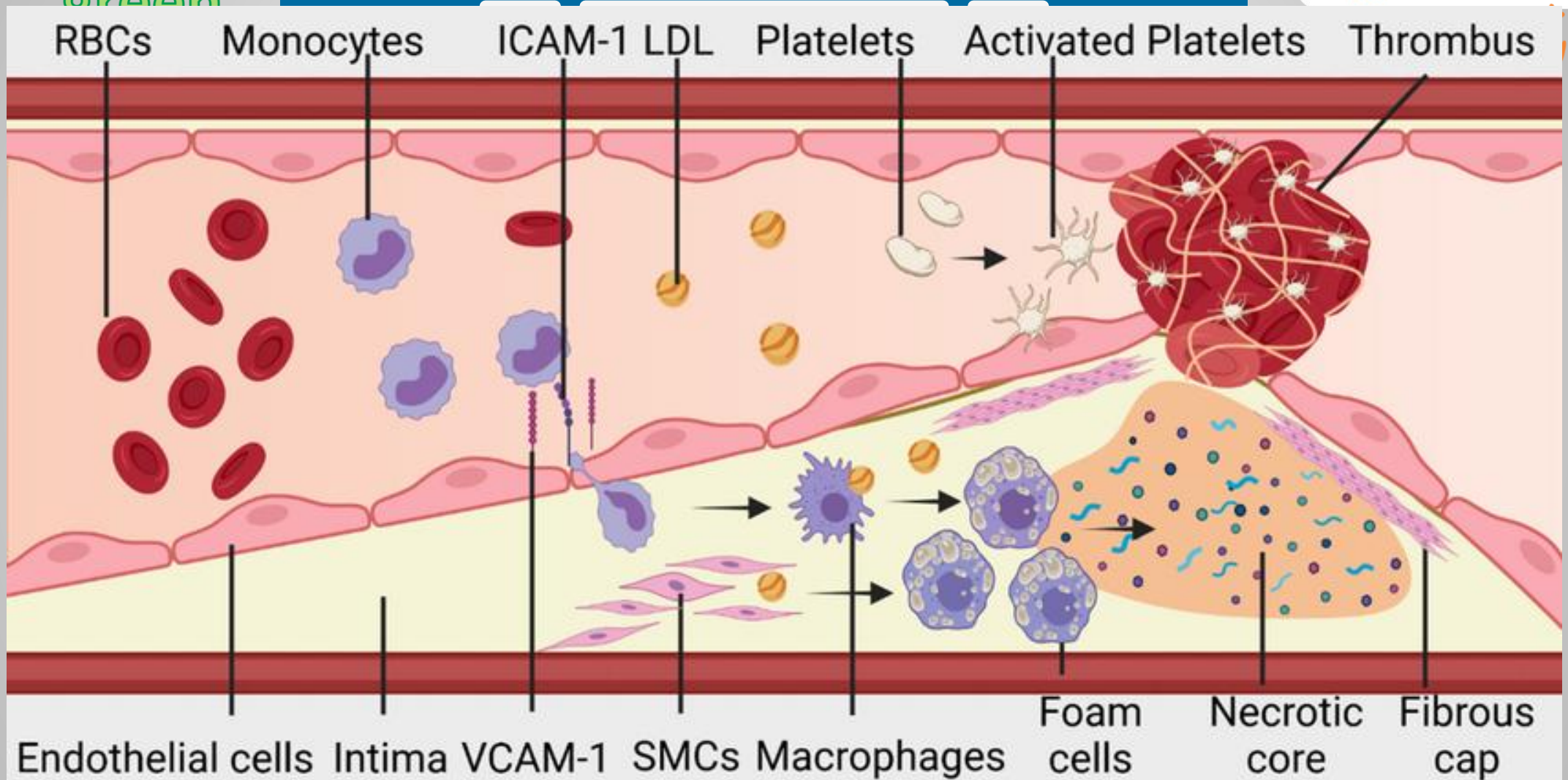


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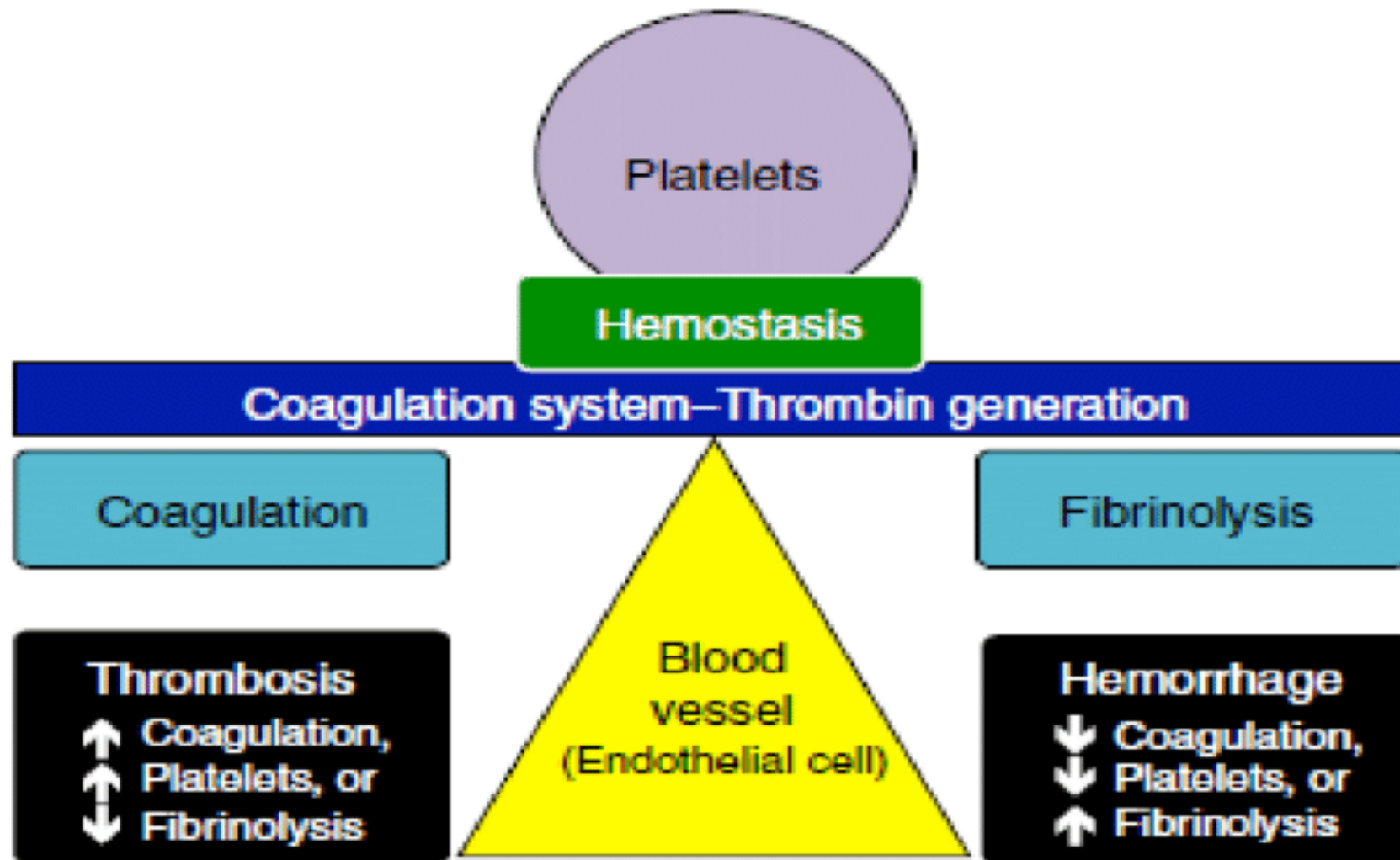


Thrombus Formation





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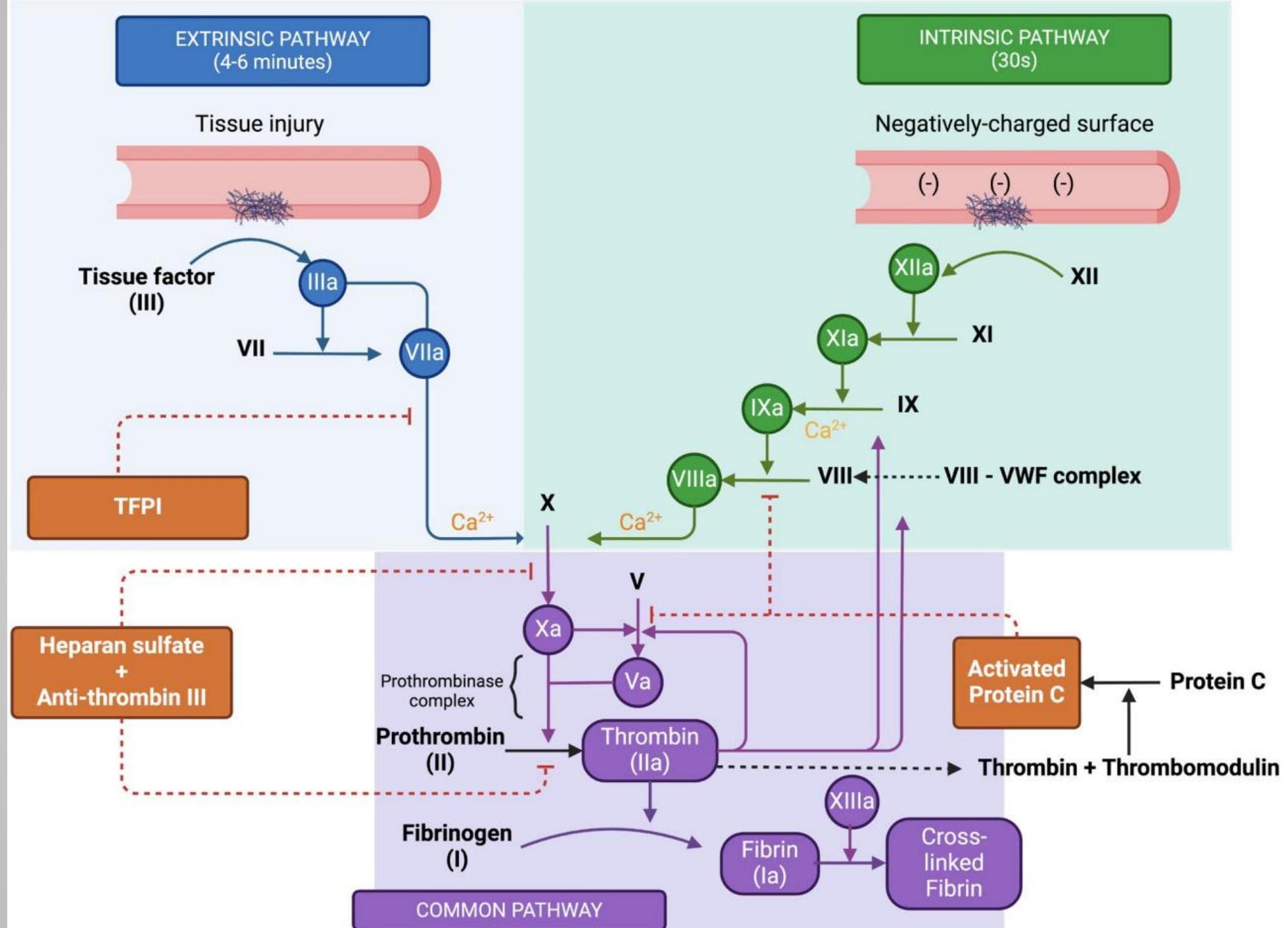
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Coagulation disorders	<i>Congenital:</i> Haemophilia A, Haemophilia B (Christmas disease) and Von Willebrand's disease. <i>Acquired:</i> Anticoagulants, liver disease, DIC, vit K deficiency.
Vascular defects	<i>Congenital:</i> Connective tissue disease (Ehlers-Danlos syndrome), Osler-Weber-Rendu syndrome. <i>Acquired:</i> Senile purpura, infections, steroids, scurvy.
Platelet defects 1. Platelet number defects 2. Platelet structural defects	Decrease marrow production: Aplastic anaemia, marrow infiltration (leukaemia, myeloma), marrow suppression (cytotoxic drugs, radiotherapy). Excessive destruction: immune thrombocytopenic purpura (ITP), SLE, CLL, heparin treatment, viruses. Thrombotic thrombocytopenic purpura (TTP), sequestration (as in hypersplenism). Myeloproliferative disease, increase urea, Von Willebrand's disease, Bernard-Soulier (giant platelet) syndrome, alcoholism, drug-induced (Aspirin, NSAID).
Fibrinolytic defects	DIC and streptokinase treatments.





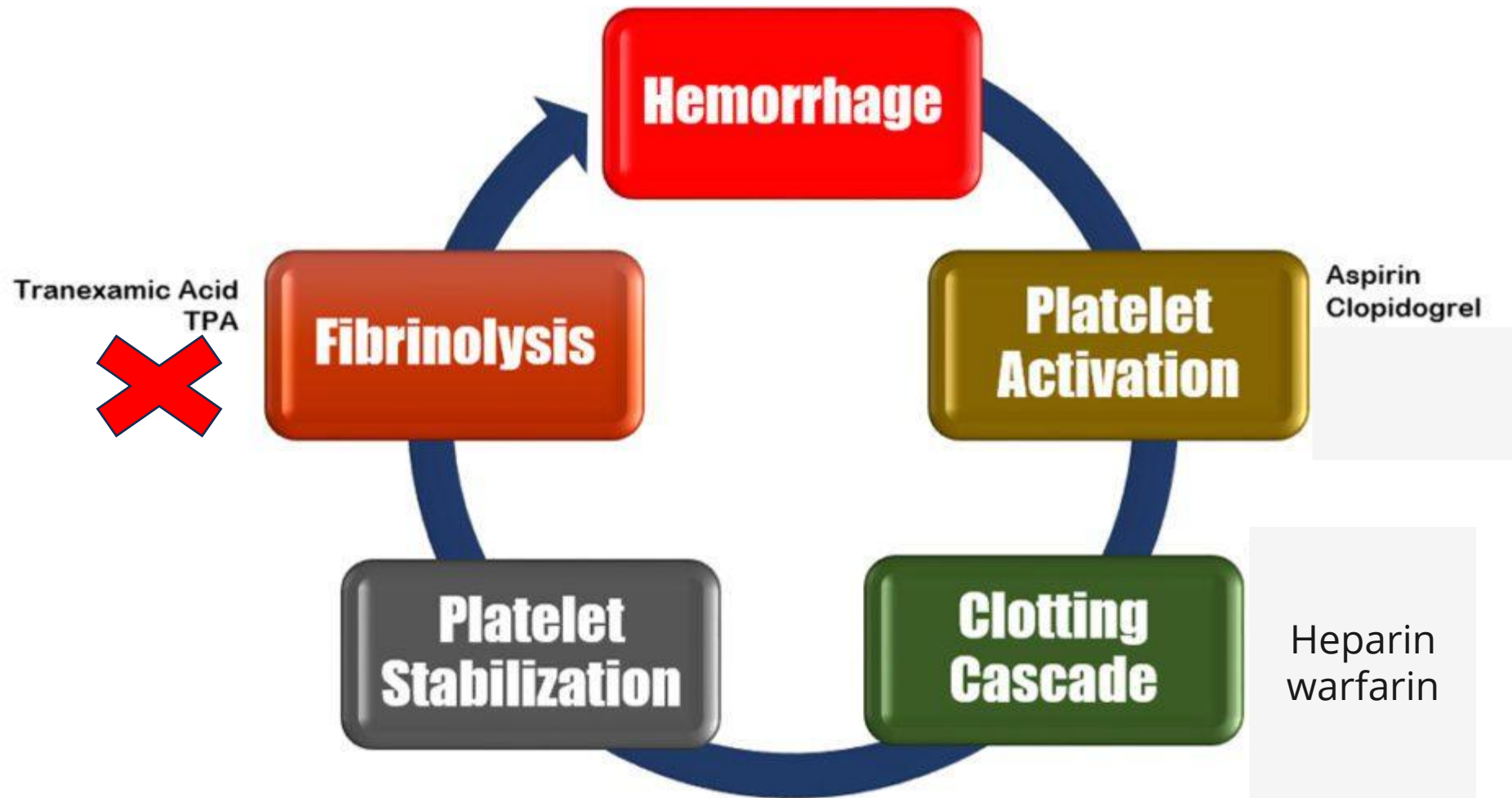


Figure 2: Anticoagulants and Antiplatelet Medications and their Effect on Coagulation



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